

SLF4J Logging Exercises

Exercise 1: Logging Error Messages and Warning Levels

Problem

Demonstrate ERROR and WARN logging.

Setup

Add SLF4J and Logback dependencies.

Java Implementation

```
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class LoggingExample{
    private static final Logger logger=
        LoggerFactory.getLogger(LoggingExample.class);

    public static void main(String[] args){
        logger.error("This is an error message");
        logger.warn("This is a warning message");
    }
}
```

Output

Program executes successfully and log messages are displayed/generated.

Exercise 2: Parameterized Logging

Problem

Demonstrate parameterized logging.

Setup

Add SLF4J and Logback dependencies.

Java Implementation

```
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class ParameterizedLogging{
    private static final Logger logger=
        LoggerFactory.getLogger(ParameterizedLogging.class);
}
```

```
public static void main(String[] args){  
    String user="John";  
    int age=25;  
    logger.info("User {} is {} years old",user,age);  
}  
}
```

Output

Program executes successfully and log messages are displayed/generated.

Exercise 3: Using Different Appenders

Problem

Log messages to both console and file.

Setup

Create logback.xml with console and file appenders.

Java Implementation

```
import org.slf4j.Logger;  
import org.slf4j.LoggerFactory;  
  
public class AppenderExample{  
    private static final Logger logger=  
        LoggerFactory.getLogger(AppenderExample.class);  
  
    public static void main(String[] args){  
        logger.debug("Debug message");  
        logger.info("Information message");  
        logger.warn("Warning message");  
        logger.error("Error message");  
    }  
}
```

Output

Program executes successfully and log messages are displayed/generated.